

CASOPS Control UI — Functional Specification v1

Item	Value
Document	<code>spec/ui.v1.md</code>
Title	CASOPS Control UI functional specification
Version	<code>ui.v1</code>
Date	2026-08-31
Host API	Public FastAPI plane <code>/api/v3</code> (<code>casops.api.control</code> , schema 3.0)
Style source	<code>spec/common-style.html</code>
UI source root	<code>ui/</code> (mandatory — see §1)
Spec	<code>common_agent_structure.md</code> (CASOPS-FS-COMMON-AGENT-STRUCTURE-V3A)
Operator guide	<code>user_guide.v1.md</code>
Status	DRAFT — UI not implemented

This document specifies a browser UI that lets an **operator** completely manage every CASOPS agent the host can load, **only** through the public control-plane API. It is not a chat wrapper and not a second control plane.

1. Source and style (normative)

1.1 All UI source lives in `ui/`

Every file that implements this UI MUST be saved under the repository `ui/` folder. That includes application code, stylesheets extracted or adapted from the style guide, static assets, tests for the UI, and UI build config.

Path	Role
<code>ui/</code>	Sole source tree for the Control UI
<code>ui/README.md</code>	How to install, run, and point the UI at a control plane
<code>ui/src/</code>	Application source (pages, components, API client, state)
<code>ui/public/</code>	Static assets that are not generated
<code>ui/tests/</code>	UI unit/contract tests
<code>spec/common-style.html</code>	App style. Read-only design system. Do not duplicate a second visual language.
<code>spec/ui.v1.md</code>	This specification (not UI source)
<code>spec/svg/</code>	Screen wireframes for this spec (not UI application source)

Forbidden:

- UI application source in `src/casops/`, `services/`, `agents/`, or repo root
- A second design system, dark-theme-only shell, or ad-hoc color palette that contradicts `spec/common-style.html`
- Calling internal service ports (`8081-8087`) from the browser. The UI talks to the **public** plane only.

The host library stays in `src/casops/`. The UI is glass over `/api/v3`, the same way Hermes dashboards sit on a gateway and OpenClaw Control UI sits on a gateway — it does not become the engine.

1.2 App style is `spec/common-style.html`

The Control UI **MUST** use `spec/common-style.html` as the app style. That file is the CASOPS design system (v2.1): light theme, Inter/system-ui, stone surfaces, indigo “common” accent, status pills that combine **text + dot + color**.

Implementers copy tokens and component recipes from that file into `ui/` (for example `ui/src/styles/` and `ui/src/components/`). They do not restyle the product as a generic dashboard.

Tokens (from `spec/common-style.html`)

Token	Class / value	Use
Base BG	<code>bg-white</code> #ffffff	Page canvas
Surface	<code>bg-stone-50</code> #fafaf9	Section wells, table chrome
Elevated	<code>bg-stone-100</code>	Disabled, stale
Border	<code>border-stone-200</code> #e7e5e4	Cards, nav, inputs
Primary text / CTA	<code>text-stone-900</code> / <code>bg-stone-900</code> #1c1917	Body titles, primary buttons
Secondary text	<code>text-stone-500</code> #78716c	Meta, timestamps
Common accent	<code>indigo-600</code> #4f46e5 + <code>indigo-50</code>	Host-owned, verified, compose hash
Pattern / router	<code>violet-600</code>	DAG / MRO
Custom fork	<code>amber-500</code>	Operator-local agent vs template
Success / live	<code>emerald-500</code>	Run ok, attestation ok
Error / failed	<code>red-500</code> / <code>red-600</code>	<code>CasopsError</code> , containment
Font	<code>Inter</code> , <code>system-ui</code> , <code>sans-serif</code>	All UI chrome
Mono	<code>font-mono</code>	<code>agent_id</code> , <code>compose_hash</code> , <code>trace_id</code> , error codes

Status pills (mandatory)

Never color alone. Use the `STATUS_CFG` set from the style file:

live, running, queued, self_refine, delayed, reconnecting, degraded, failed, recovery, complete, unavailable, stale, cancelled.

Map host facts:

Host fact	Pill
GET /health ok and last /api/v3 success < 5s	live
Runtime run in flight	running
Consolidate queued: true	queued
Improvement candidate evaluated, not approved	self_refine
Last success 30s–5m	delayed
Fetch retry	reconnecting
HTTP 5xx or containment_required	degraded / recovery
containment_stop set or IMP_CORRIGIBILITY	failed + recovery banner
baseline_safe run finished, containment_stop: null	complete
Control plane unreachable	unavailable
Projection older than 5m	stale — disable mutations
Operator cancelled	cancelled

Stale projection: primary mutation buttons disabled (**Stale – Refresh First** ghost/disabled pattern from the style file).

Components to reuse as recipes

From spec/common-style.html, the UI ships equivalents of:

- Sticky top nav (h-14, border-b border-stone-200, bg-white/95 backdrop-blur-md)
- Indigo 7×7 rounded mark + wordmark **caso**
- StatusPill, CommonBadge (schema 3.0, compose hash prefix)
- Registry **agent cards** (white, rounded-2xl, indigo icon well, 3-stat grid)
- Fleet **swarm cards** (status + compact meta)
- Canvas **graph nodes** (indigo solid = host-composed; amber dashed = local/unverified)
- Primary CTA: rounded-full bg-stone-900 text-white
- Host/common CTA: rounded-full bg-indigo-600 text-white
- Destructive: rounded-full bg-red-600 text-white
- Outlined secondary: border-stone-200 bg-white
- Cards: bg-white rounded-2xl border border-stone-200 hover:shadow-md

Layout: max-w-6xl for marketing-style pages; **ops shell** uses full width with a left nav (still the same tokens).
Mobile: stack nav, one-column cards.

2. Product intent

2.1 Job to be done

An operator can, without CLI or raw curl:

- 1. See every agent the host can load.
- 2. Inspect structure, composition, capabilities, protocols, cache, memory, plugins, safety, improvement, validation, and corrigibility.
- 3. Compose-preview and run `baseline_safe`.
- 4. Inspect traces, replay (including counterfactual), root-cause, and evidence graphs.
- 5. Query / write-candidate / delete / verify memory with tenant+subject, and enqueue consolidation.
- 6. Validate plugins without executing them.
- 7. Evaluate / approve / rollback improvement candidates under the mutation contract.
- 8. See eval honesty (`NOT_RUN`, `INDICATIVE`, `MEASURED_LOCAL`) without mistaking it for a production pass.

2.2 What this UI is not

- Not a Hermes-style embedded TUI chat or xterm session. CASOPS `baseline_safe` has no streaming chat API.
- Not an OpenClaw “ask the agent to build widgets” board. Widgets would be agent-authored UI; CASOPS forbids the agent from owning the control plane.
- Not a second public API. No UI-only backend that mutates agent folders behind `/api/v3`.
- Not a production-activation console. `production_activation_requested` stays false; the UI never offers “go live”.

2.3 Research: Hermes, OpenClaw, and what we take

Reviewed:

- **Hermes Agent web dashboard / Control Interface** — Status, Chat, Config, Sessions, Agents (profile CRUD + per-agent dashboard / memory / skills / cron), Office swarm monitor, Usage, MCP, password gate. Form-driven config instead of YAML. Auto-refresh status.
- **Hermes Studio** — crews, approvals, memory graph, skills, cron. Explicitly “not a chat wrapper”.
- **OpenClaw Control UI** — Overview, Channels, Sessions, Cron, Chat, Skills, Config, Debug, Logs. Agent hierarchy. Dashboard/chat split for widgets.
- **OpenClaw Command Center / Mission Control** — fleet graph, agent drawer (identity, tools, files), kanban tasks, spend, human-in-the-loop approvals, audit.

Adopt

Pattern	From	CASOPS use
Fleet home + agent switcher	Hermes, OpenClaw	Agent list → agent workspace
Per-agent tabbed workspace	Hermes agent detail	Structure, compose, run, memory, plugins, safety, improvement, eval
Mutation / approval as first-class UI	Hermes Studio, OpenClaw	Header strip: actor, reason, parent, dry-run

Pattern	From	CASOPS use
Memory browser with scope	Hermes Memory tab	Tenant + subject required; empty other-tenant is success
Live vs stale honesty	<code>common-style.html</code> + OpenClaw overview	Status pills, disable stale mutations
Trace / session inspector	OpenClaw sessions, Hermes sessions	Trace, replay, evidence
Kanban for governed work	OpenClaw / Hermes Office	Improvement candidates: proposed → evaluated → approved / rejected / rolled back
Local-only default	Hermes dashboard on localhost	UI defaults to <code>http://127.0.0.1:8080</code>

Reject

Pattern	Why
Chat-as-the-product	No chat completions API on <code>/api/v3</code>
YAML/config.yaml editors for 150 fields	Agent contract is a folder ; compose-preview is the merge UI
Agent-built dashboard widgets	Agent must not own chrome or gates
Channel/Telegram/Discord binding	Not in CASOPS v3 public plane
Cron job manager	No cron API on <code>/api/v3</code>
Token spend analytics as a primary nav	No usage meter API; do not invent cost charts
Dark-only admin skins	Style guide is light stone/indigo

3. Host API the UI consumes

3.1 Public plane

Base URL (operator-configurable, default `http://127.0.0.1:8080`).

All mutating calls (`POST`, `PUT`, `PATCH`, `DELETE`) under `/api/v3` **MUST** send:

Header	UI control	Notes
<code>x-casops-actor</code>	Actor select	<code>human_operator</code> <code>independent_approver</code> <code>host_service</code> . Never default <code>agent_runtime</code> .
<code>x-casops-reason</code>	Required text	Placeholder: "why this change"
<code>x-casops-expected-parent</code>	Text, default <code>none</code>	

Header	UI control	Notes
<code>x-casops-dry-run</code>	Toggle, default on	<code>true</code> / <code>false</code>

GET does not send the contract. If any mutation header is missing, the host returns non-200 `IMP_UNSIGNED`. The UI surfaces that as a blocking toast and does not retry silently.

`agent_runtime` on approve → `IMP_SELF_APPROVAL`. The UI must not offer that actor on Approve.

3.2 Route catalog (every spec §19 path has a screen)

`{id}` = selected `agent_id`. `{tid}` = `root_trace_id`. `{aid}` = artifact id. `{mid}` = memory id. `{cid}` = candidate id.

Method	Path	UI surface
GET	<code>/health</code>	Shell connection pill (not a v3 resource; still used)
GET	<code>/api/v3/agents/{id}/structure</code>	Fleet card + Structure tab
GET	<code>/api/v3/agents/{id}/resolved</code>	Structure / Compose "resolved" panel
POST	<code>/api/v3/agents/{id}/compose-preview</code>	Compose tab primary action
GET	<code>/api/v3/agents/{id}/runtime/plan</code>	Run tab DAG
GET	<code>/api/v3/agents/{id}/runtime/capabilities</code>	Capabilities tab
GET	<code>/api/v3/agents/{id}/capabilities/matrix</code>	Capabilities tab
POST	<code>/api/v3/agents/{id}/capabilities/verify</code>	Capabilities "Verify"
GET	<code>/api/v3/agents/{id}/runtime/context-budget</code>	Run tab / Context
GET	<code>/api/v3/agents/{id}/cache/stats</code>	Cache panel
POST	<code>/api/v3/agents/{id}/cache/invalidate</code>	Cache "Invalidate"
GET	<code>/api/v3/agents/{id}/protocols</code>	Protocols tab
GET	<code>/api/v3/agents/{id}/plugins</code>	Plugins tab
POST	<code>/api/v3/agents/{id}/plugins/validate</code>	Plugins "Validate without exec"
GET	<code>/api/v3/agents/{id}/memory/policy</code>	Memory tab header
GET	<code>/api/v3/agents/{id}/memory/hierarchy</code>	Memory tab
POST	<code>/api/v3/agents/{id}/memory/query</code>	Memory search
POST	<code>/api/v3/agents/{id}/memory/write-candidate</code>	Memory write (disabled if policy <code>none</code>)

Method	Path	UI surface
POST	/api/v3/agents/{id}/memory/consolidate	Memory “Enqueue consolidate”
DELETE	/api/v3/agents/{id}/memory/{mid}	Memory delete
POST	/api/v3/agents/{id}/memory/{mid}/verify-deletion	Memory verify
GET	/api/v3/traces/{tid}	Trace inspector
POST	/api/v3/traces/{tid}/replay	Replay
POST	/api/v3/traces/{tid}/replay?counterfactual=	Counterfactual replay
GET	/api/v3/traces/{tid}/root-cause	Root-cause panel
GET	/api/v3/artifacts/{aid}/evidence-graph	Evidence panel
GET	/api/v3/agents/{id}/safety/incidents	Safety tab
POST	/api/v3/agents/{id}/safety/redteam	Safety “Run red-team fixture”
GET	/api/v3/agents/{id}/improvement/candidates	Improvement board
POST	/api/v3/agents/{id}/improvement/candidates/{cid}/evaluate	Evaluate
POST	/api/v3/agents/{id}/improvement/candidates/{cid}/approve	Approve (independent_approver or host_service only)
POST	/api/v3/agents/{id}/improvement/rollback/{version}	Rollback
GET	/api/v3/agents/{id}/improvement/ledger	Ledger table
GET	/api/v3/agents/{id}/regression/suite	Eval / regression
GET	/api/v3/agents/{id}/corrigibility/attestation	Attestation banner on every agent page
GET	/api/v3/agents/{id}/validation/report	Validation tab
POST	/api/v3/agents/{id}/runtime/run	Run tab primary action

Query parameters the UI must expose (not bury):

Call	Params
memory query	tenant, subject (and optional text if the client sends it)
memory write-candidate	tenant, subject, text
memory delete / verify-deletion	tenant, subject required
replay	optional counterfactual

3.3 Discovery gap (must be explicit)

Today's OpenAPI **does not** include `GET /api/v3/agents` (list). Complete fleet management still needs a list of `agent_ids`.

UI-v1 required companion (host addition, public plane only):

```
GET /api/v3/agents
```

Response (illustrative):

```
{
  "agents": [
    {
      "agent_id": "casops.template.baseline_safe",
      "folder": "agents/_template_v3",
      "structure_id": "casops.common_agent.v3",
      "schema_version": "3.0",
      "role": "BaselineSafeTemplate"
    }
  ]
}
```

Until that route exists, the UI **Settings** → **Known agents** lets the operator paste IDs. The UI then probes `GET .../structure` per ID. Empty list is an empty state, not fake agents.

The UI **MUST NOT** scan the operator's disk from the browser.

3.4 Error envelope

```
{
  "error": {
    "code": "IMP_UNSIGNED",
    "message": "The request was rejected by host policy.",
    "containment_required": false
  }
}
```

UI: toast + inline banner. If `containment_required: true`, full-width **Recovery** bar (`STATUS_CFG.recovery`), mutations other than "Reload attestation" disabled.

Show `error.code` in `font-mono`. Never show only a generic "Request failed".

4. Information architecture

```
Shell
└─ Connection (base URL, health pill, actor strip)
```

└ Fleet	/	GET list or known IDs +
/structure		
└ Agent workspace	/agents/:agentId	
└└ Overview	attestation, structure summary, last run	
└└ Structure	/structure, /resolved	
└└ Compose	POST /compose-preview	
└└ Run	/runtime/plan, POST /runtime/run, context-budget,	
cache		
└ Trace	/traces/{id}, replay, root-cause, evidence-graph	
└ Capabilities	/capabilities/matrix, POST verify,	
/runtime/capabilities		
└ Protocols	GET /protocols	
└ Memory	policy, hierarchy, query, write, delete, verify,	
consolidate		
└ Plugins	GET /plugins, POST validate	
└ Cache	GET stats, POST invalidate	
└ Safety	incidents, POST redteam	
└ Improvement	candidates, evaluate, approve, rollback, ledger	
└ Validation	/validation/report, /regression/suite	
└└ Corrigibility	/corrigibility/attestation	
Settings	/settings	base URL, actor defaults,
known agent IDs		

Deep links are shareable. Example: /agents/casops.template.baseline_safe/run.

5. Shell

5.1 Frame

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Control UI

casops.template.baseline_safe ▾

Live

as_of just now

Fleet

Overview

Structure

Compose

Run

Trace

Capabilities

Protocols

Memory

Plugins

Cache

Safety

Improvement

Validation

Corrigibility

Settings

Shell frame

Sticky nav, agent switcher, Live pill, actor strip. Content area is a placeholder.

This frame is shared by every agent screen.

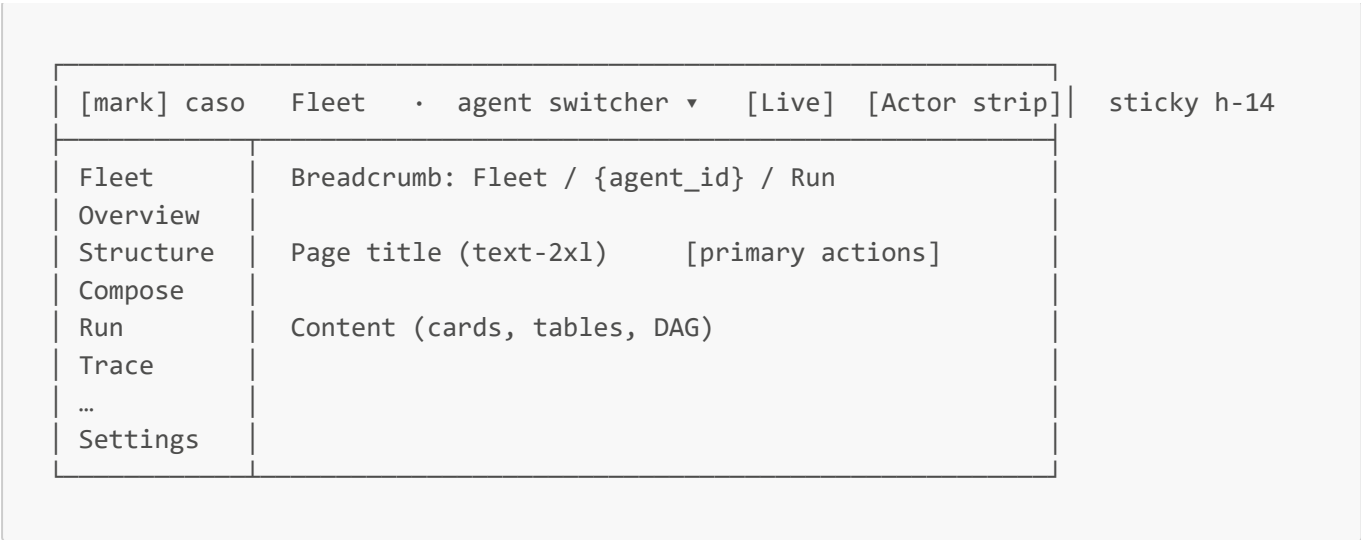
Top: caso mark · agent switcher · Live pill · as_of

Strip: actor · reason · expected-parent · dry-run ON (default)

Left: 16 destinations with indigo rail on the selected item

Stale projection disables primary mutations (Stale — Refresh First).

agent_runtime hides Approve. Empty reason disables POST/DELETE.



- Left nav: stone-50 well, 14px labels, indigo mark on the selected item.
- Agent switcher in the top bar (Hermes profile switcher): current `agent_id` mono, search, jump.
- Connection pill: `Live` / `Unavailable` / `Reconnecting` from `/health` + last v3 result.
- **Actor strip** (always visible on mutation pages): actor `<select>`, reason `<input>`, expected-parent `<input>`, dry-run `<toggle>` default ON. A lock icon when dry-run is on.

5.2 Actor strip rules

- Stored in session (not localStorage for actor if the machine is shared; Settings may opt-in).
- Dry-run ON: primary buttons labeled **Preview** / **Run (dry-run)** even when the host still executes some POSTs (compose-preview writes no locks regardless).
- Switching actor to `agent_runtime` hides Approve and shows a red note: "Agent identities cannot approve."
- Reason empty → mutation buttons disabled in the UI (fail closed before the network).

5.3 Global empty / error

State	UI
No control plane	<code>unavailable</code> card: "Start unicorn on :8080" + Settings link
No agents	Empty fleet: "Add a known agent_id or implement GET /api/v3/agents"
<code>INH_PARENT_MISSING</code>	"Unknown agent_id" on the workspace
Containment	Recovery banner; attestation panel forced open

6. Screens

Each screen lists **purpose**, **layout**, **API**, **states**, **actions**. Primary buttons use stone-900 or indigo-600 per §1.2.

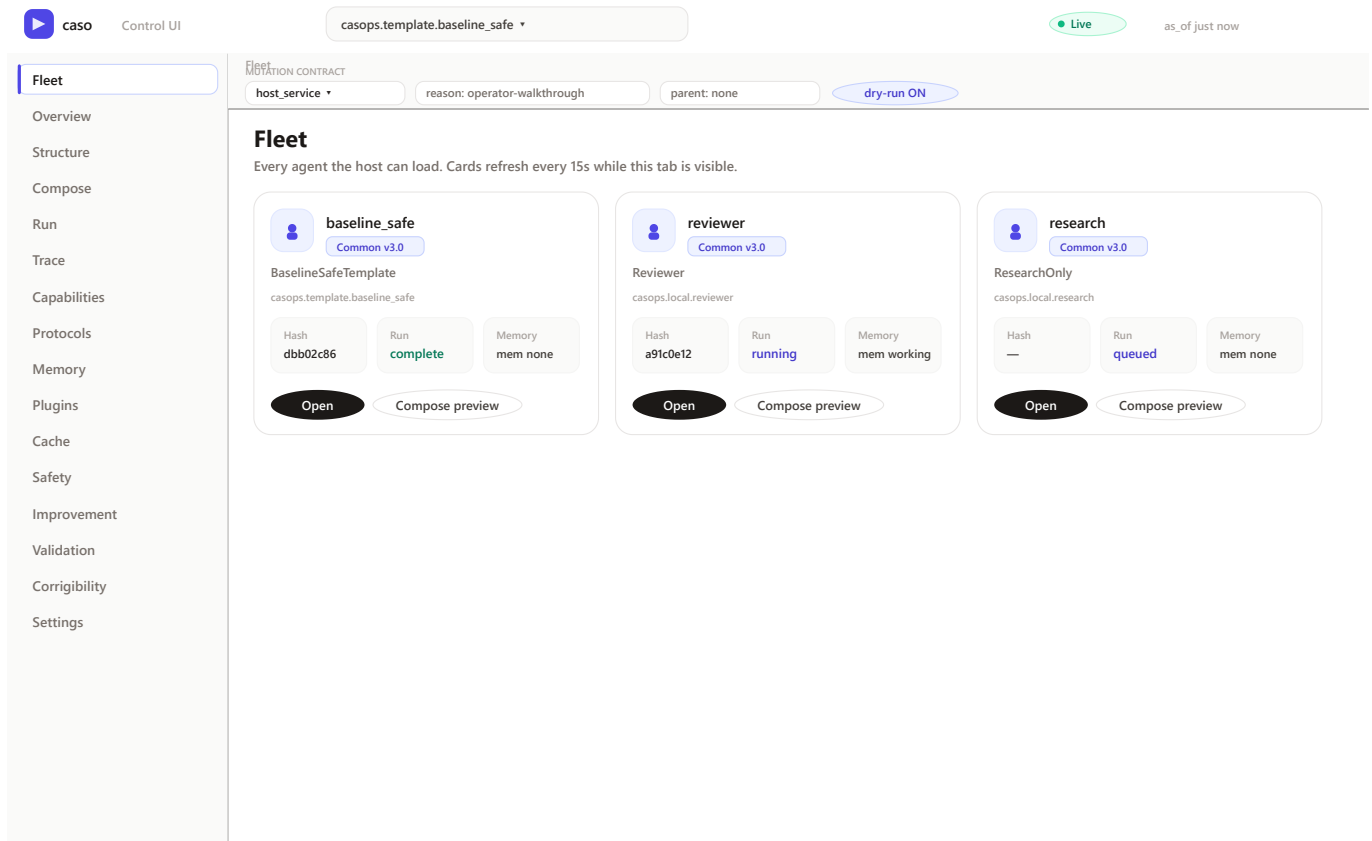
Wireframes live in `spec/svg/` and use tokens from `spec/common-style.html`.

Screen	Figure
Shell	<code>spec/svg/00-shell.svg</code>
Fleet	<code>spec/svg/01-fleet.svg</code>

Screen	Figure
Overview	spec/svg/02-overview.svg
Structure	spec/svg/03-structure.svg
Compose	spec/svg/04-compose.svg
Run	spec/svg/05-run.svg
Trace	spec/svg/06-trace.svg
Capabilities	spec/svg/07-capabilities.svg
Protocols	spec/svg/08-protocols.svg
Memory	spec/svg/09-memory.svg
Plugins	spec/svg/10-plugins.svg
Cache	spec/svg/11-cache.svg
Safety	spec/svg/12-safety.svg
Improvement	spec/svg/13-improvement.svg
Validation	spec/svg/14-validation.svg
Corrigibility	spec/svg/15-corrigibility.svg
Settings	spec/svg/16-settings.svg

6.1 Fleet (/)

Purpose. Single pane of all agents (OpenClaw overview + Hermes Home, without chat/cron).



Layout. Page title “Fleet”. Grid of **registry agent cards**:

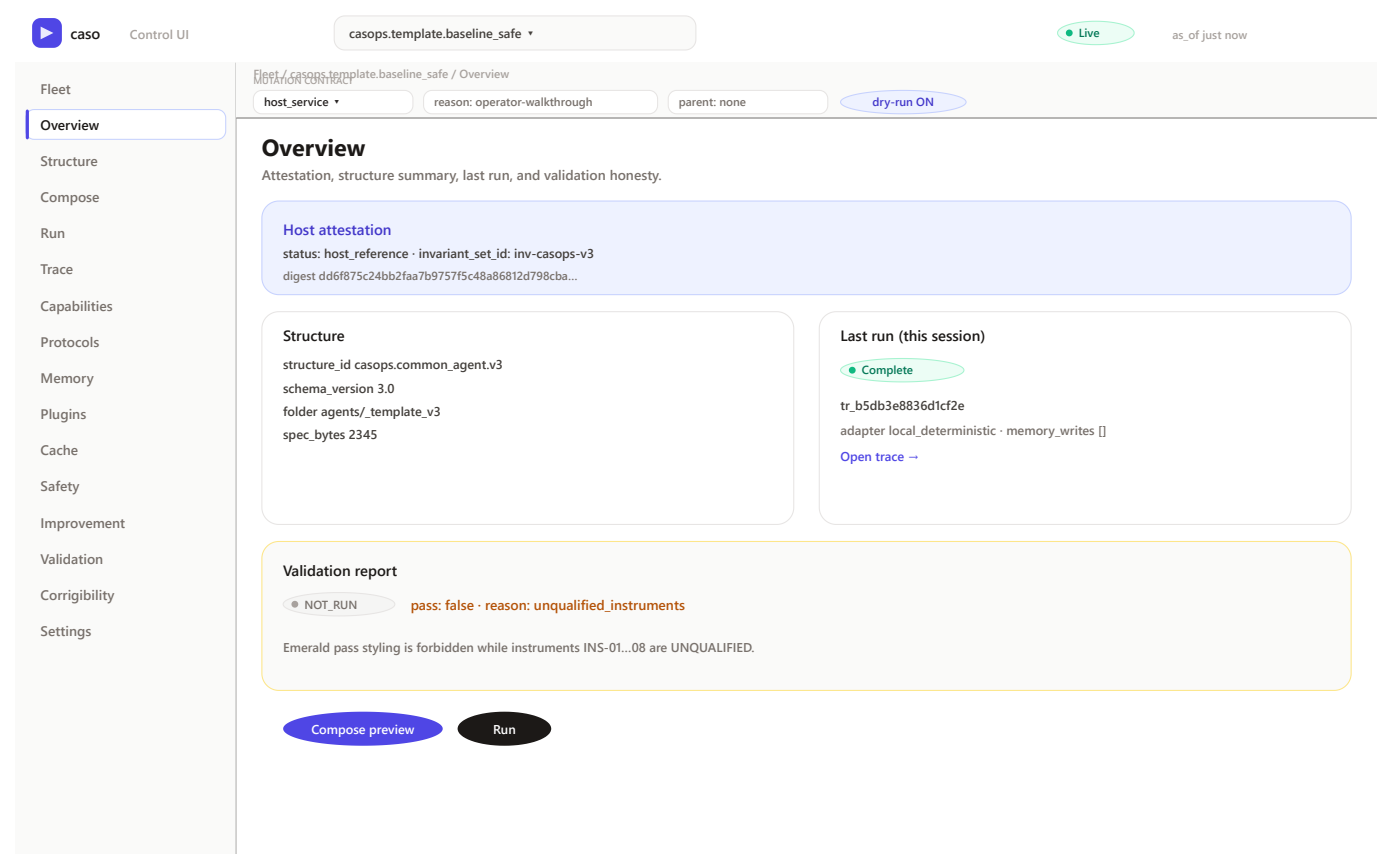
- Bot icon in indigo well
- **agent_id** (semibold) + **CommonBadge** schema 3.0
- Role / folder (secondary text)
- Stats: last compose_hash prefix (mono), last run status pill, memory mode
- CTA: **Open** (stone-900). Secondary: **Compose preview** (outlined)

API. `GET /api/v3/agents` if present; else known IDs × `GET .../structure`. Optionally `GET .../corrigibility/attestation` and `GET .../memory/policy` for card meta (parallel, tolerate failure).

Refresh. Every 15s while the tab is visible; pause when stale/hidden.

6.2 Agent overview (`/agents/:id`)

Purpose. Attestation + “can I run this?”



Layout.

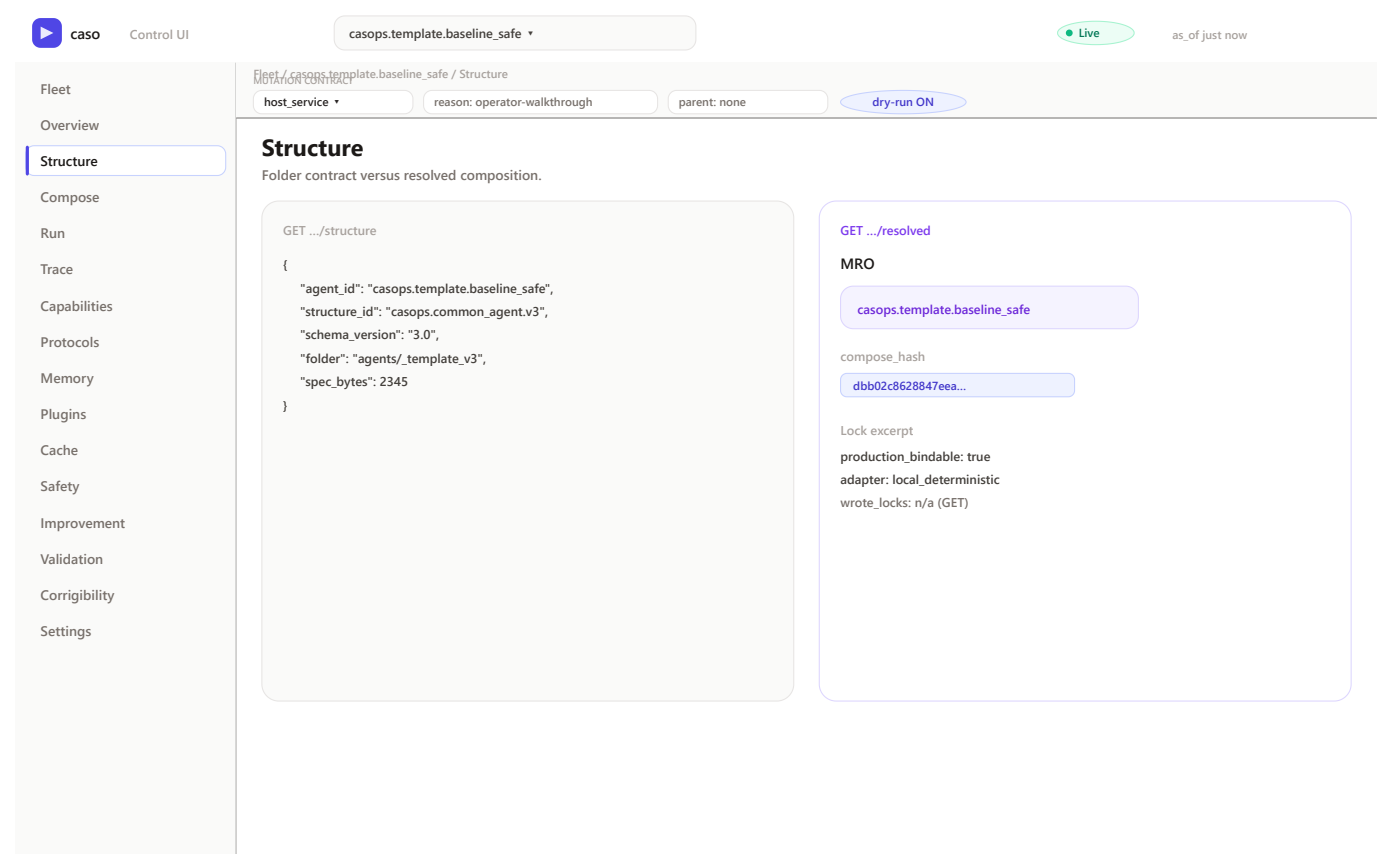
- 1. Recovery/attestation card (indigo border if ok, red if containment).
- 2. Structure summary: `structure_id`, `schema_version`, folder path, spec bytes.
- 3. Last run snapshot if the client cached a `root_trace_id` this session.
- 4. Validation honesty: `GET .../validation/report` — if `pass: false` / `NOT_RUN`, emerald **must not** be used. Use `stale` or `unavailable` pill plus the `reason` string.

API. `GET .../structure`, `GET .../corrigibility/attestation`, `GET .../validation/report`.

Actions. **Compose preview**, **Run** (both jump to those tabs with actor strip filled).

6.3 Structure (`/agents/:id/structure`)

Purpose. Folder contract vs resolved composition.



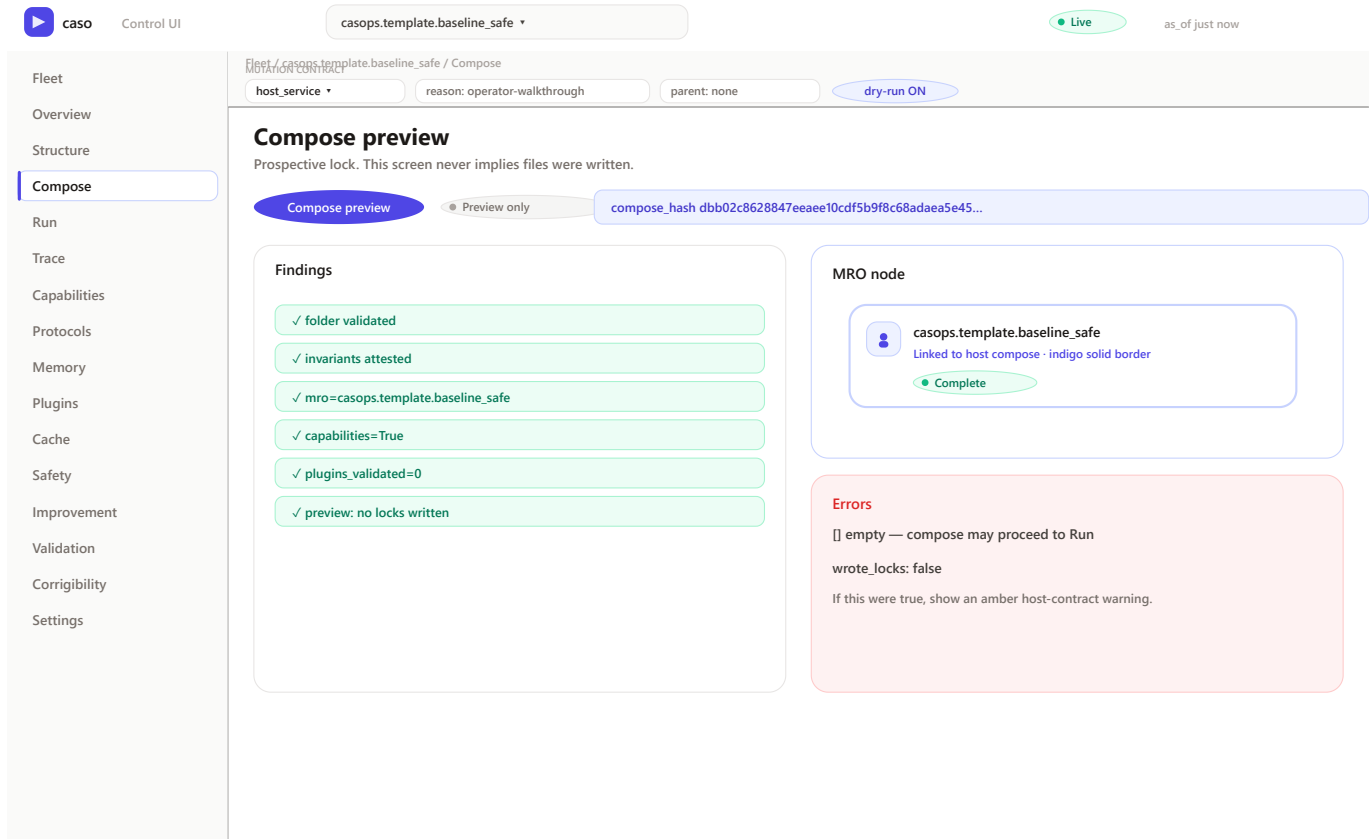
Layout. Two columns (stack on mobile):

- Left: raw structure JSON (read-only, mono, stone-50 well).
- Right: resolved MRO list (violet), **compose_hash**, lock excerpt.

API. **GET .../structure**, **GET .../resolved**.

6.4 Compose (**/agents/:id/compose**)

Purpose. Prospective lock. **Never** implies files were written.



Layout.

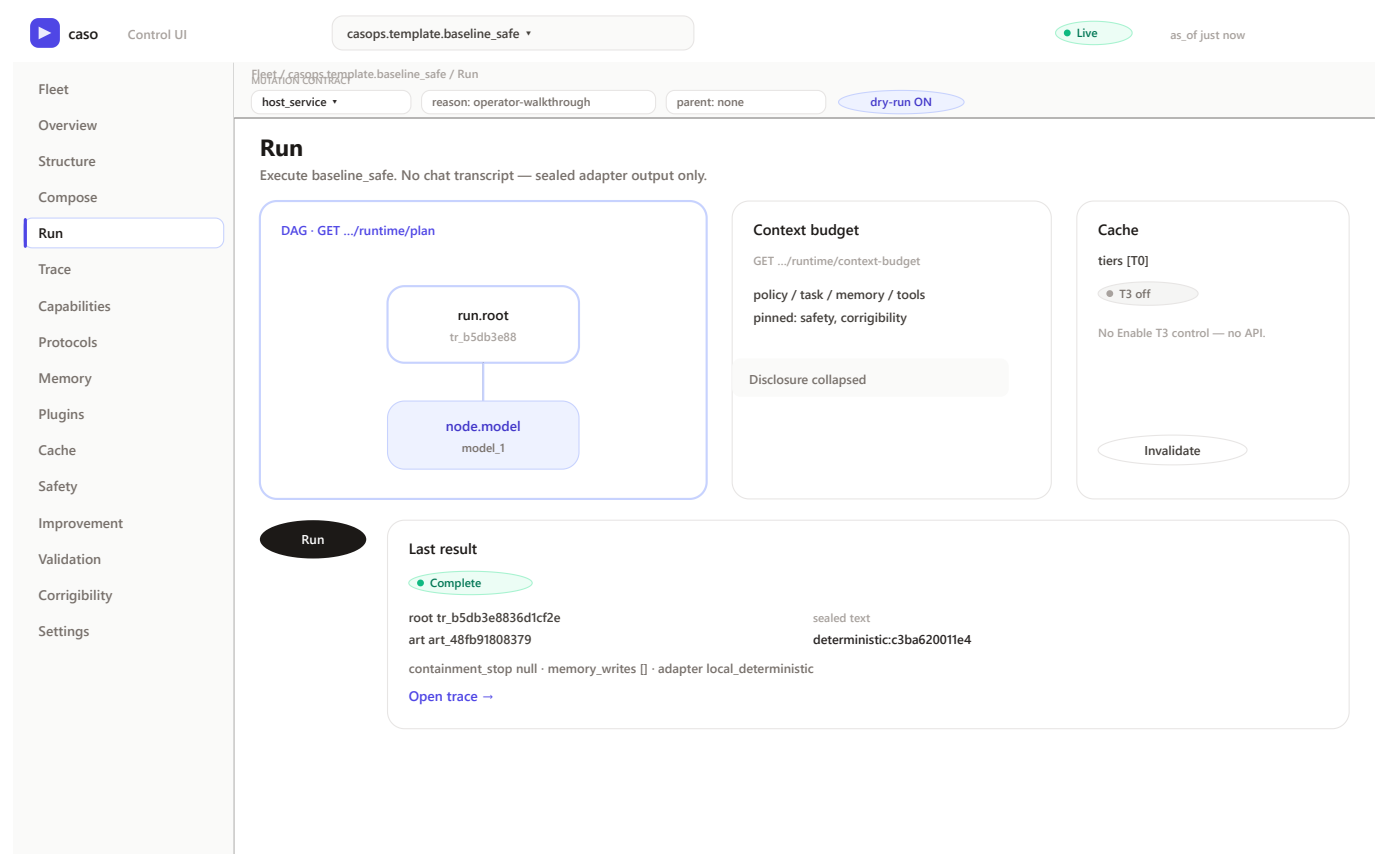
- Findings as a checklist (emerald check / red X).
- `compose_hash` in `CommonBadge` / mono.
- `wrote_locks` MUST render as a badge: `false` → outlined stone "Preview only"; if ever `true`, amber warning (host bug relative to v1 contract).
- Errors list in red-50 card.
- Optional DAG of MRO (canvas node recipe, indigo solid).

API. `POST .../compose-preview` (mutation headers).

Action. Primary **Compose preview**. Disable when actor strip invalid or projection stale.

6.5 Run (`/agents/:id/run`)

Purpose. Execute `baseline_safe` and show the DAG.



Layout.

- DAG from GET .../runtime/plan (canvas nodes; model node indigo).
- Context budget JSON in a disclosure (GET .../runtime/context-budget).
- Cache stats (GET .../cache/stats): tiers, t3_enabled (if false, stone badge "T3 off").
- Invalidate cache outlined button → POST .../cache/invalidate.
- Primary Run → POST .../runtime/run.
- Result card: root_trace_id, artifact.id, containment_stop, memory_writes, adapter, cancelled. Link Open trace.

Honesty. Template has no streaming tokens. Do not draw a fake chat transcript. Show the sealed text/digest from the run JSON only.

6.6 Trace (/agents/:id/traces/:tid and /traces/:tid)

Purpose. Inspect, replay, counterfactual, evidence (OpenClaw session detail, without chat).

The screenshot shows the 'caso' Control UI. The sidebar on the left lists various system components, with 'Trace' currently selected. The main panel displays a detailed trace for the component 'casops.template.baseline_safe'. At the top, there's a header with the component name and a 'Live' indicator. Below this, a 'Trace' section provides a summary: 'One root span. Replay and counterfactual are mutations. No memory-write control.' The 'Span tree' shows a root span 'tr_b5db3e8836d1cf2e' with a 'Complete' status, which contains a child span 'sp_3ffcae3a2e482f78'. At the bottom of the span tree are three buttons: 'Replay', 'counterfactual: route', and 'Counterfactual'. To the right of the span tree, there are two informational panels. The 'Root cause' panel shows the cause as 'none' and the adapter as 'local_deterministic'. The 'Evidence graph' panel shows a claim with support 'deterministic_adapter' and a note that it is unsupported.

Layout.

- Span tree (one root). Root pill **complete** / **failed**.
- Replay: **Replay** (POST `.../replay`), **Counterfactual** opens a field for `counterfactual=` then POST.
- Root-cause panel: GET `.../root-cause`.
- Evidence: GET `.../artifacts/{aid}/evidence-graph` when artifact id is known (from run result or evidence payload).

API. As in §3.2. Replay is a mutation (headers required). Counterfactual must not offer a “write memory” control.

6.7 Capabilities (`/agents/:id/capabilities`)

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Live

as, of just now

Fleet

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Fleet / casops.template.baseline_safe / Capabilities

host_service reason: operator-walkthrough parent: none dry-run ON

Capabilities

Unverified claims stay amber. Never paint them green.

Verify

POST .../capabilities/verify

Capability	Status	Production bindable
model.local_deterministic	VERIFIED	yes
model.batch_invariant_kernels	UNVERIFIED	no — amber, not green
plugin.wasm.il	UNVERIFIED	no — amber, not green

Disclosure: GET .../runtime/capabilities (same matrix, raw JSON)

Layout. Matrix table: capability, status (VERIFIED → emerald badge). **Verify** → POST .../capabilities/verify. Also show GET .../runtime/capabilities in a disclosure.

Unverified production-bindable claims: amber, not green.

6.8 Protocols (/agents/:id/protocols)

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Fleet / casops.template.baseline_safe / Protocols

host_service reason: operator-walkthrough parent: none dry-run ON

Protocols

Read-only. There is no PATCH on the v3 plane.

GET .../protocols

```
{
  "agent_id": "casops.template.baseline_safe",
  "protocols": "{ \"schema_version\": \"3.0\", ... }",
  "pinned": [\"agent_message\", \"event\"]
}
```

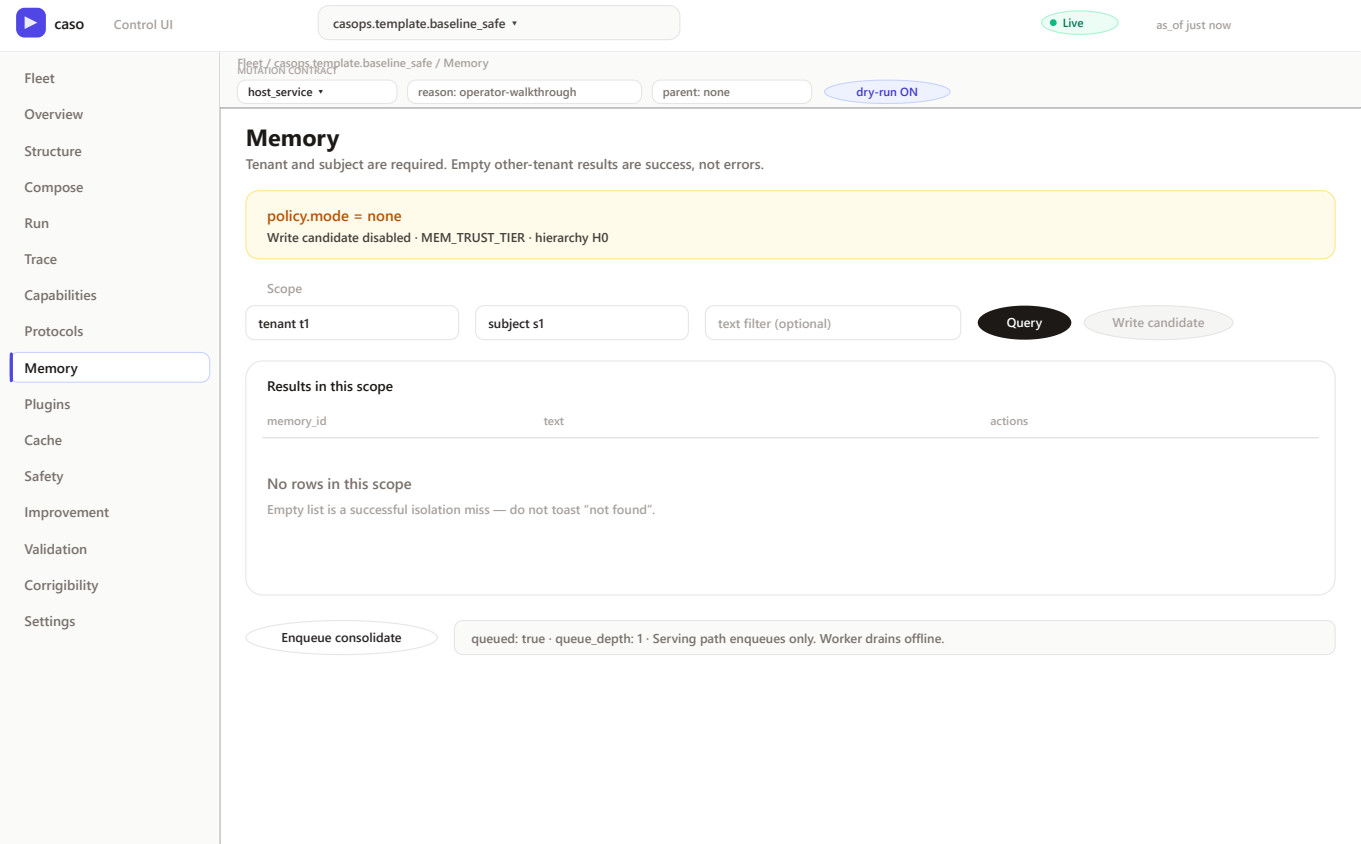
No edit, import, or save control on this screen.

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Read-only JSON from `GET .../protocols`. Mono well. No edit (no PATCH in v3).

6.9 Memory (`/agents/:id/memory`)

Purpose. Scoped memory ops (Hermes Memory tab, with CASOPS isolation).

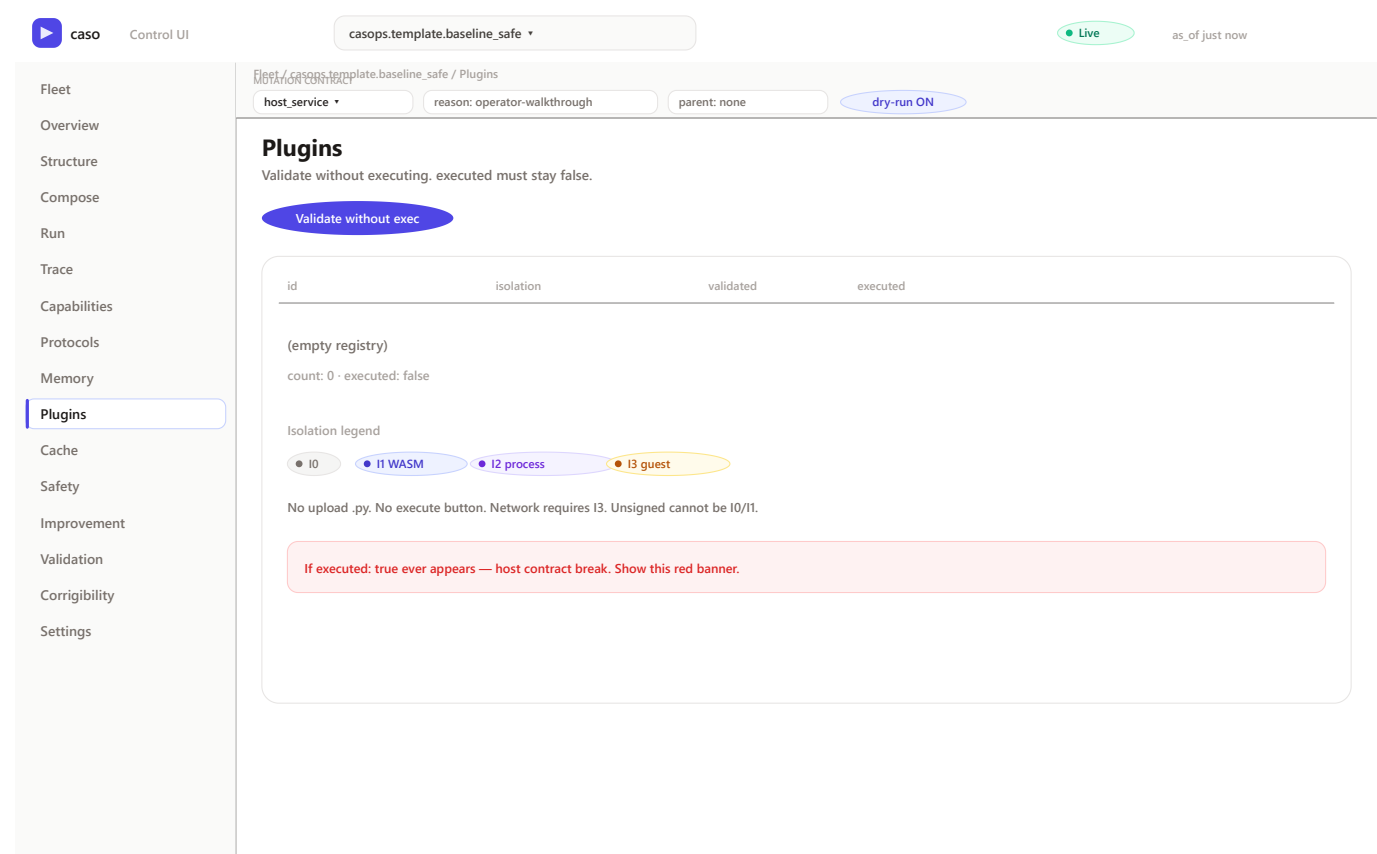


Layout.

- Policy + hierarchy header. If `mode` is `none` / `disabled`, **Write candidate** is disabled with tooltip `MEM_TRUST_TIER`.
- Scope bar: **tenant** and **subject** inputs (required). Persist last pair per agent in session.
- Search: `POST .../query?tenant=&subject=` (optional text).
- Results table: `memory_id`, text excerpt. Actions: **Delete**, **Verify deletion**.
- Delete/verify send the same tenant+subject. Wrong scope → banner `MEM_SCOPE`, do not toast “not found”.
- Cross-tenant query returning `[]` is a **successful empty** state (“No rows in this scope”), not an error.
- **Enqueue consolidate** → `POST .../consolidate`. Show `queued` and `queue_depth`. Copy: “Serving path enqueues only. The consolidation worker drains offline.”

6.10 Plugins (`/agents/:id/plugins`)

Purpose. Validate without exec.



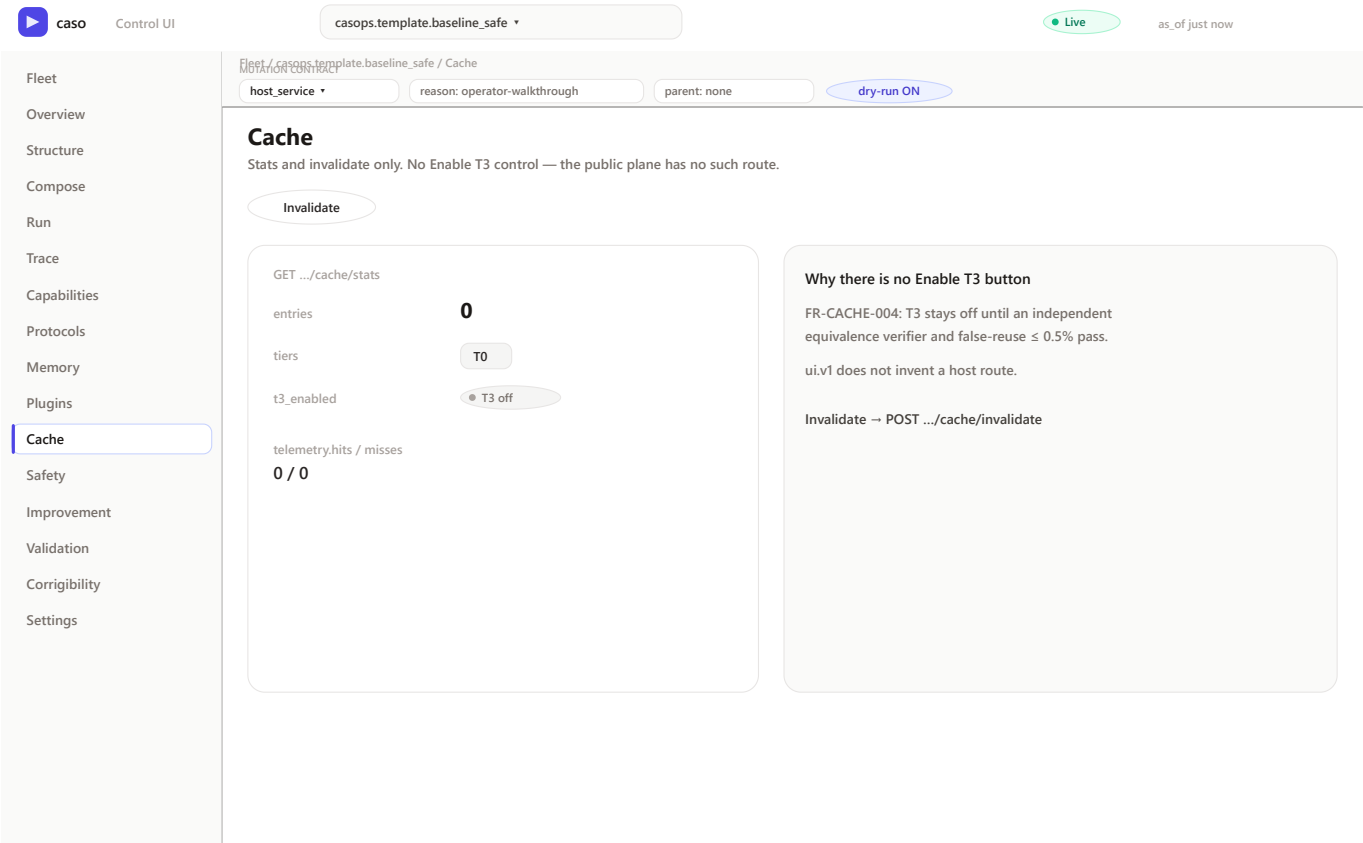
Layout. Table of plugins (**id**, isolation, **validated**, **executed** must display **false**). **Validate** → **POST** **.../plugins/validate**.

If **executed: true** ever appears, red banner (host contract break).

Isolation badges: I0 stone, I1 indigo, I2 violet, I3 amber. Tooltip: I3 needs sandbox; network requires I3; unsigned cannot be I0/I1.

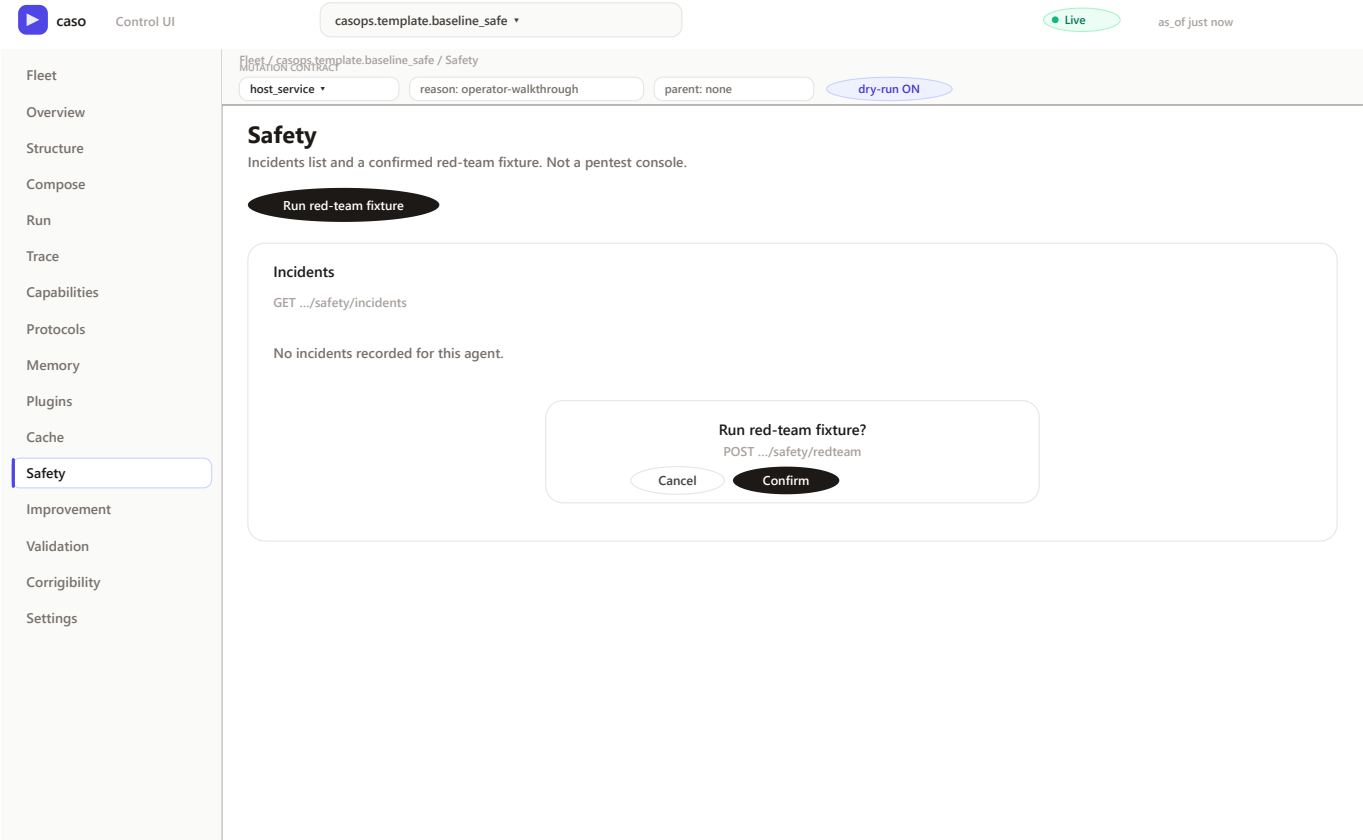
No “upload .py plugin” control. No execute button in v1 (no execute route on the public plane except as part of a run DAG).

6.11 Cache (**/agents/:id/cache**)



Maybe a panel on Run or its own tab. Stats JSON + **Invalidate**. If **t3_enabled** is false, do not show a fake "Enable T3" that has no **/api/v3** route.

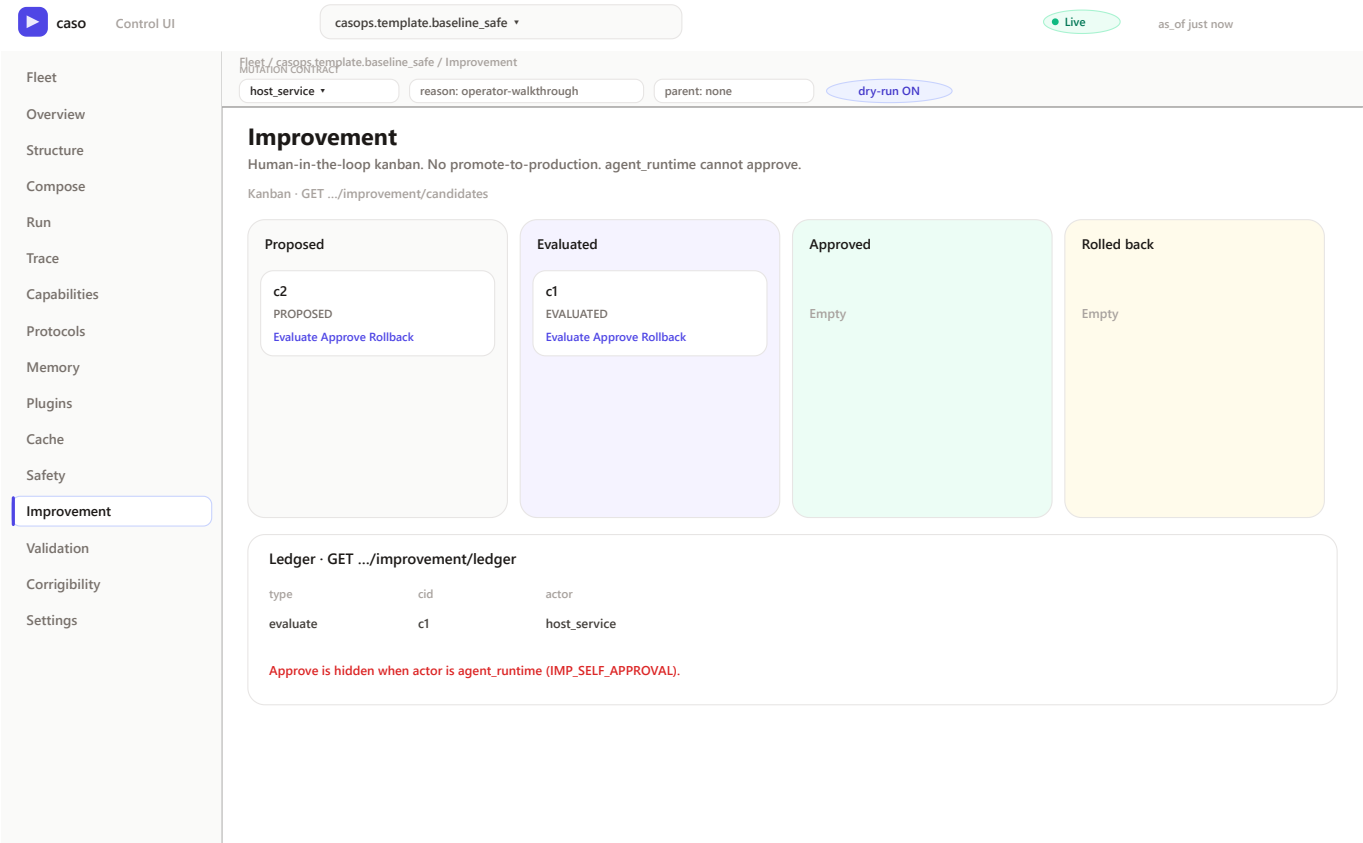
6.12 Safety (**/agents/:id/safety**)



Incident list. **Run red-team fixture** → **POST .../redteam**. Confirm dialog (outlined cancel, stone-900 confirm).

6.13 Improvement (/agents/:id/improvement)

Purpose. OpenClaw-style human-in-the-loop, Hermes-style approvals — mapped to candidates/ledger.



Layout. Kanban (stone-50 board):

Column	Candidate state
Proposed	missing / PROPOSED
Evaluated	EVALUATED
Approved	HUMAN_APPROVED
Rolled back	after rollback action

Cards: **id**, agent_id, state. Actions on the card:

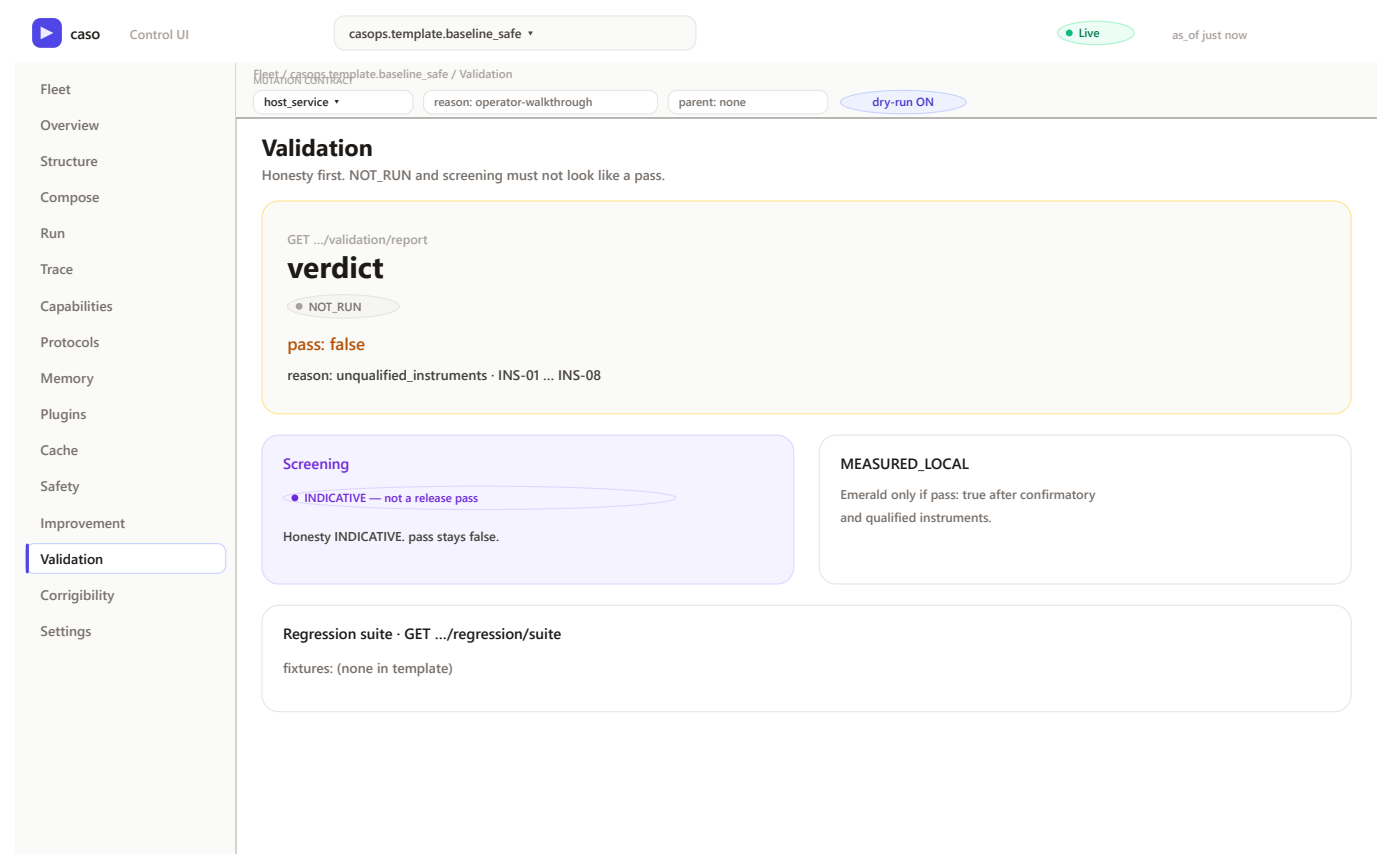
- **Evaluate** → POST .../candidates/{cid}/evaluate
- **Approve** → only if actor is independent_approver or host_service; confirm modal restates actor + reason
- **Rollback** → version field + POST .../rollback/{version}

Ledger table below: GET .../ledger.

No “promote to production” button.

6.14 Validation (/agents/:id/validation)

Purpose. Honesty-first eval (IQ-01).

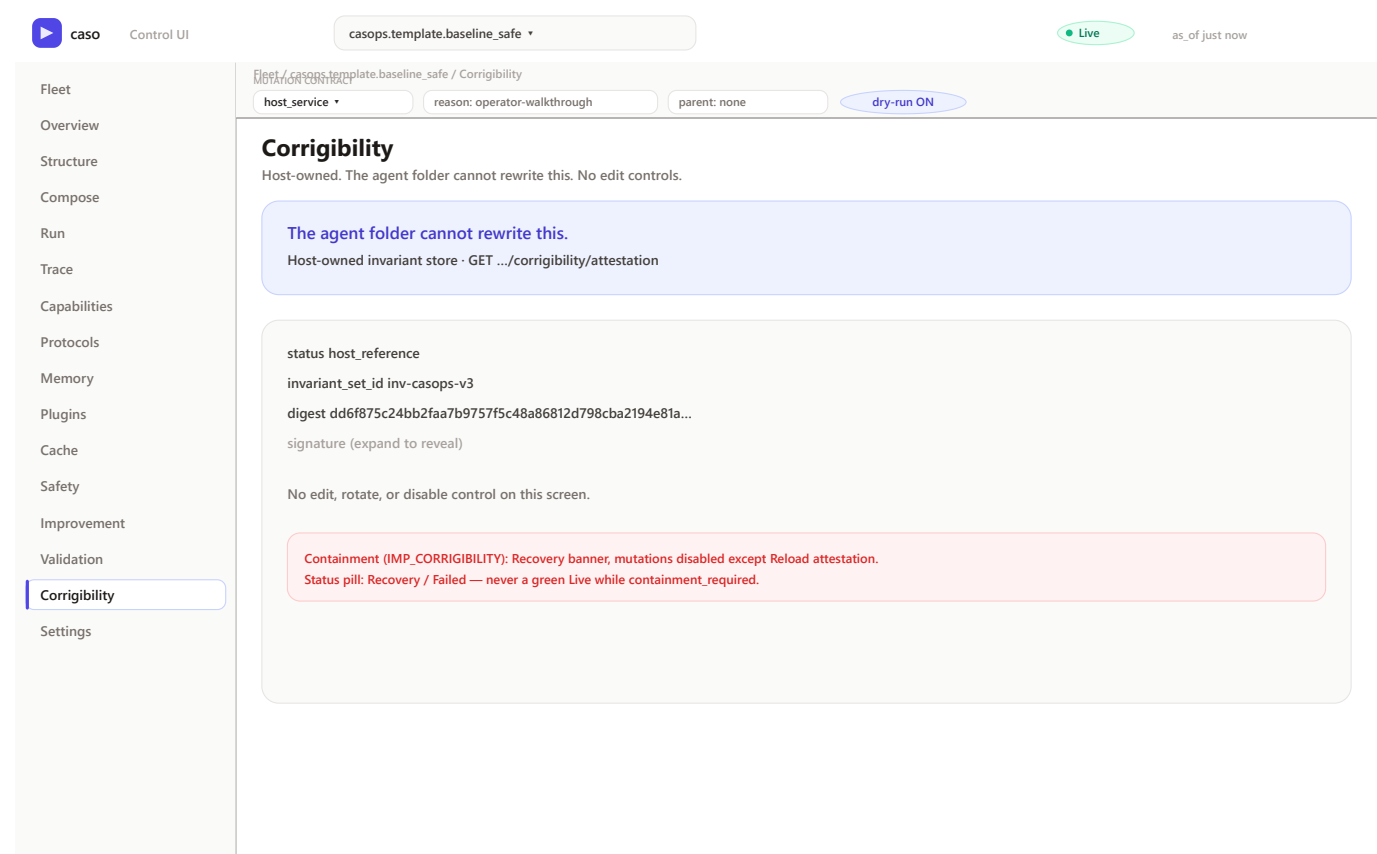


Layout.

- Report from `GET .../validation/report`.
- If `verdict` is `NOT_RUN` or `pass` is false: large stone/amber panel, **not** a green check. Show `reason` and instrument list.
- `INDICATIVE` screening: violet pill “Screening — not a release pass”.
- `MEASURED_LOCAL`: emerald only if `pass: true`.
- Regression suite names: `GET .../regression/suite`.

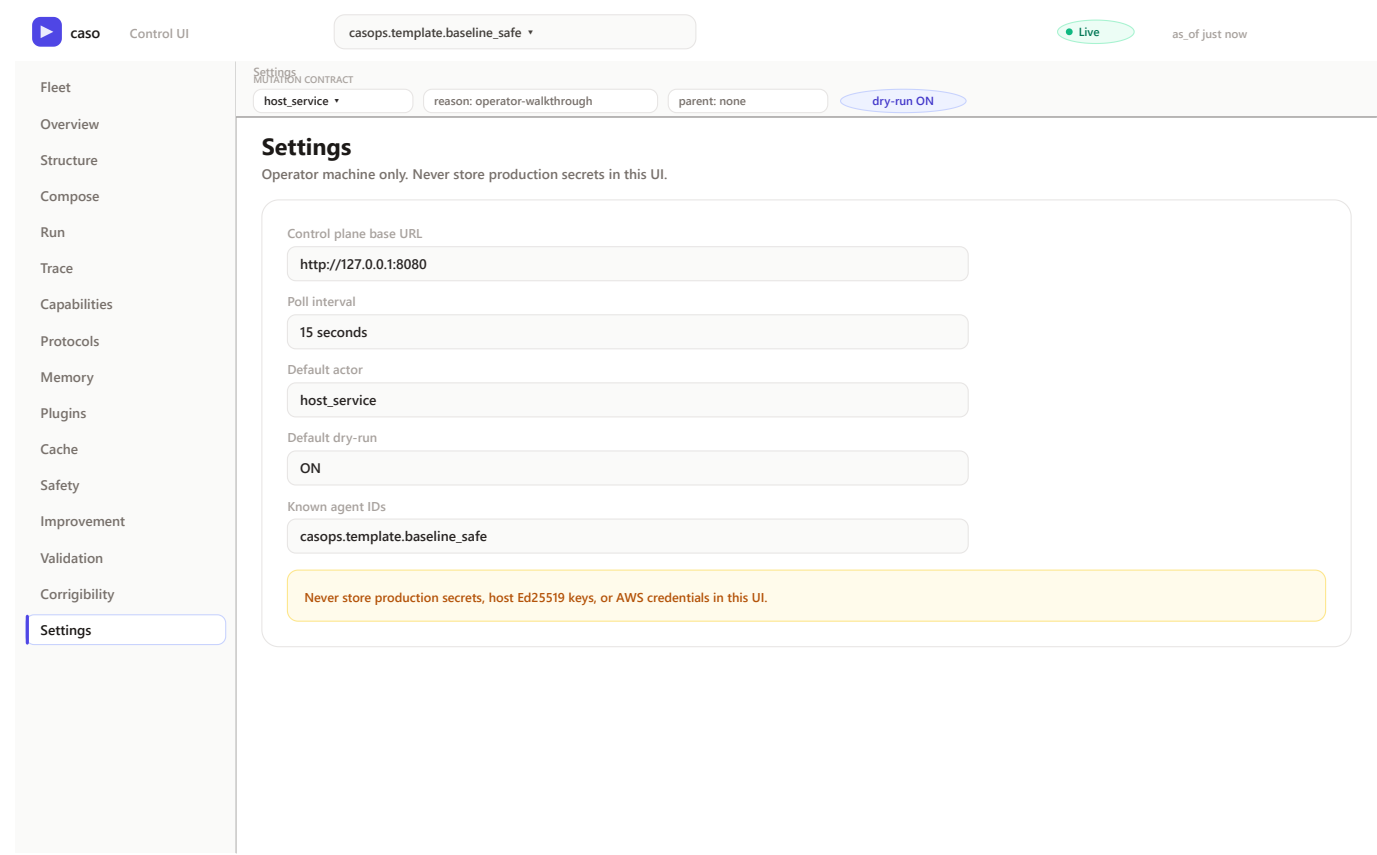
Do not embed `casops-eval` CLI. Optional: link to `user_guide.v1.md` §13.

6.15 Corrigibility (`/agents/:id/corrigibility`)



Attestation JSON: digest, signature, status, invariant_set_id. Mono. Host-owned callout: “The agent folder cannot rewrite this.” No edit controls.

6.16 Settings (/settings)



- Control plane base URL

- Poll interval
 - Default actor / dry-run
 - Known agent IDs (until list API exists)
 - "Never store production secrets in this UI" note
-

7. Cross-cutting behavior

7.1 Client

- TypeScript SPA is preferred (matches Hermes web). Bundle from `ui/`.
- One API module: `ui/src/api/v3.ts` — injects mutation headers on POST/DELETE.
- No raw `fetch` from random components without that module.
- Timeouts: 30s default; run/compose 120s.

7.2 Freshness

- `as_of` timestamp on every loaded panel (style file: `text-sm text-stone-500`).
- After a mutation, re-GET the panels that mutation affects (do not trust the POST body alone when a GET exists).
- SSE is **not** in v3. Do not fake a live token stream. Polling is honest.

7.3 Dry-run

Default ON. Compose-preview already writes no locks. Run still executes the deterministic adapter; the UI label must not say "no side effects" for Run. Memory delete in dry-run: if the host still tombstones, the UI must show the real result (do not hide `tombstoned: true`).

7.4 Accessibility

- Status never color-only (style system).
- Buttons have visible focus rings.
- Actor strip fields are labelled.
- Keyboard: skip to main, agent switcher typeahead.
- Contrast: stone-900 on white; do not use indigo-50 text on white for primary copy.

7.5 Responsive

- `< md`: left nav becomes a top drawer; actor strip stacks; cards one column (style file `grid-cols-1 sm:grid-cols-2 lg:grid-cols-3`).

7.6 Security

- UI is an operator tool on localhost by default.
 - Do not put host Ed25519 keys in the browser.
 - Do not display full signatures as copy-paste secrets in screenshots; allow expand-to-reveal.
 - CSP: prefer same-origin API; no random third-party analytics.
-

8. Coverage matrix (complete management)

A UI implementation is **complete for v1** only if every row is true.

Capability	Screen	API	Done when
List / open agents	Fleet	list companion or known IDs + structure	Operator can open every loaded agent
See folder contract	Structure	GET structure, resolved	MRO + hashes visible
Compose without writing locks	Compose	POST compose-preview	wrote_locks shown false
Run baseline_safe	Run	POST runtime/run	One root trace displayed
Inspect / replay traces	Trace	GET/POST traces, evidence	Counterfactual query exposed
Capabilities	Capabilities	GET matrix, POST verify	Unverified not green
Protocols	Protocols	GET protocols	Read-only
Memory lifecycle	Memory	query/write/delete/verify/consolidate	Scope required; policy none disables write
Plugins validate	Plugins	GET plugins, POST validate	executed: false shown
Cache	Cache / Run	GET stats, POST invalidate	T3-off visible
Safety	Safety	GET incidents, POST redteam	Confirm before redteam
Improvement	Improvement	candidates/evaluate/approve/rollback/ledger	agent_runtime cannot approve
Validation honesty	Validation	GET report, regression	NOT_RUN not rendered as pass
Corrigibility	Overview + tab	GET attestation	Containment banner
Mutation contract	Shell	headers	Unsigned POST never looks successful
Style	all	—	spec/common-style.html tokens only

Capability	Screen	API	Done when
Source	all	—	files under ui/ only

9. Implementation notes for **ui/**

Suggested tree (normative location, informative names):

```
ui/
  README.md
  package.json           # or equivalent; UI build lives here
  src/
    main.tsx
    styles/
      tokens.css         # extracted from spec/common-style.html
    components/
      StatusPill.tsx
      CommonBadge.tsx
      ActorStrip.tsx
      AgentCard.tsx
      RecoveryBanner.tsx
    api/
      v3.ts              # all /api/v3 calls
      types.ts
    pages/
      Fleet.tsx
      AgentOverview.tsx
      ...
    shell/
      AppShell.tsx
    tests/
      api-contract.test.ts  # every SPEC_V3 path has a client function
      mutation-headers.test.ts
```

Build output may be **ui/dist/** (generated, not hand-edited).

Python host does not serve the UI in v1 unless a later change adds a static mount. Running the UI is documented in **ui/README.md** (dev server + `VITE_CASOPS_BASE=http://127.0.0.1:8080` or equivalent).

10. Out of scope for ui.v1

- Chat / completions / streaming tokens
- Creating agent folders on disk from the browser (no create-agent API)
- Editing **agent_spec.json** files in place (no PATCH folder API)
- Enabling T3 without a host route
- L5 promote-to-production
- Firecracker console
- Citation auditor UI (CLI remains **casops-citation**; optional later)

- Instrument qualification UI (no public qualify route on §19)
 - Dark theme
 - Multiplayer / SSO (localhost operator)
-

11. Acceptance

ui.v1 is accepted when:

1. All UI source is under `ui/`.
2. Visual language matches `spec/common-style.html` (light stone, indigo common, status pills with text+dot+color).
3. Every §3.2 route is reachable from a labelled control (no dead API).
4. Mutation headers are impossible to omit from the client module.
5. `NOT_RUN` / screening cannot be styled as a green pass.
6. Template walkthrough from `user_guide.v1.md` §§6–11 can be performed entirely in the UI.

End of `spec/ui.v1.md`.